

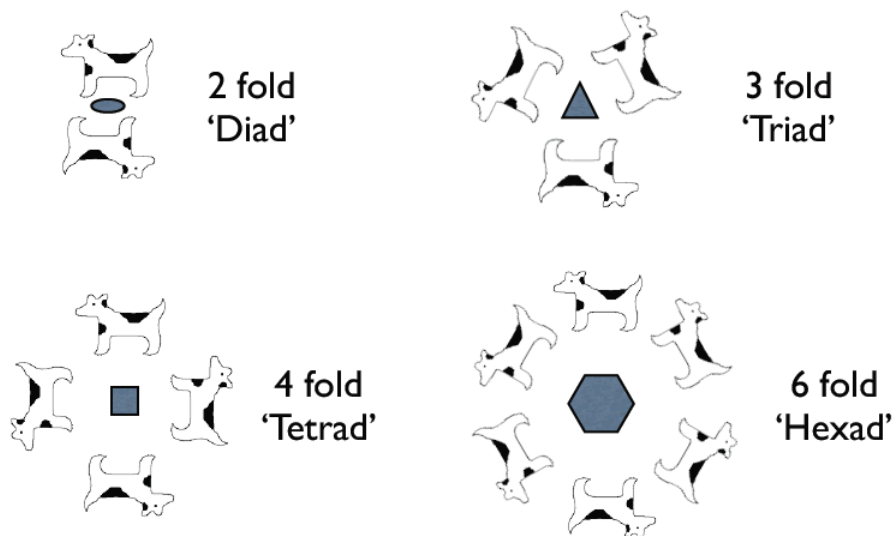
1.7.10: Symmetry

We have already met the concept symmetry in relation to crystal structures: the lattice generates the translational symmetry—the motif is repeated on every lattice point.

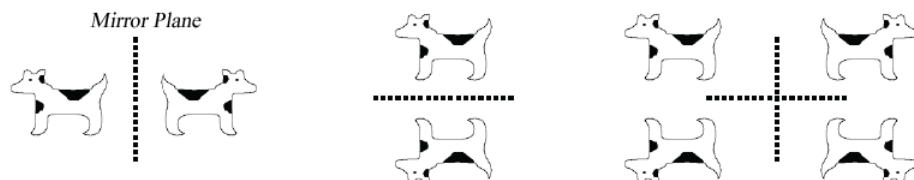
Other types of symmetry exist, including:

- rotation axes
- mirror planes
- centre of symmetry
- inversion axes (combination of rotation and centre of symmetry operations)

An n fold rotational symmetry operation rotates an object by $360^\circ/n$. Only $n = 1, 2, 3, 4,$ and 6 are permitted in a periodic lattice



An object has mirror symmetry if reflection of the object in a plane brings it into coincidence with itself:



Some objects have special symmetry about an origin such that, for any point at position x, y, z , there is an exactly similar point at $-x, -y, -z$. The origin is called a centre of symmetry ("inversion centre"). Such an object is said to be centrosymmetric:

An n-fold inversion axis is a combination of a rotation by $360/n$ followed by a centre of symmetry operation. An example of a 4-fold inversion axis is shown in the following animation:

This page titled [1.7.10: Symmetry](#) is shared under a [CC BY-NC-SA](#) license and was authored, remixed, and/or curated by [Dissemination of IT for the Promotion of Materials Science \(DoITPoMS\)](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.

- **15.7: Symmetry** by [Dissemination of IT for the Promotion of Materials Science \(DoITPoMS\)](#) is licensed [CC BY-NC-SA 2.0](#). Original source: <https://www.doitpoms.ac.uk>.